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Part A

$$\begin{aligned} 1. a) \quad R_T &= R_1 + R_2 + R_3 \\ &= 20 + 30 + 50 \\ &= 100 \Omega \\ b) \quad R_T &= \frac{1}{\frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}} \\ &= \frac{1}{\frac{1}{20} + \frac{1}{30} + \frac{1}{50}} \\ &= \frac{1}{0.05 + 0.01 + 0.02} \\ &= 12.5 \Omega \end{aligned}$$

2. Total Current

$$\begin{aligned} a) \quad I_T &= \frac{V}{R_T} \\ &= \frac{125}{100} \\ &= 1.25 A \\ b) \quad I_T &= \frac{V}{R_T} \\ &= \frac{125}{12.5} \\ &= 10 A \end{aligned}$$

3. Current Through Each Resistor

$$a) \quad I_1 = I_2 = I_3 = 1.25 A$$

$$b) \quad I_1 = I_2 = I_3 = 10 A$$

$$\begin{aligned} 4. a) \quad V_1 &= I \times R_1 = 1.25 A \times 20 = 25 V \\ V_2 &= I \times R_2 = 1.25 A \times 30 = 37.5 V \\ V_3 &= I \times R_3 = 1.25 A \times 50 = 62.5 V \\ 25 + 37.5 + 62.5 &= 125 V \end{aligned}$$

$$b) \quad I_1 = \frac{V}{R_1} = \frac{125}{20} = 6.25 A$$

$$I_2 = \frac{V}{R_2} = \frac{125}{30} = 4.17 A$$

$$I_3 = \frac{V}{R_3} = \frac{125}{50} = 2.5 A$$

$$\begin{aligned} 6.25 A + 4.17 A + 2.5 A \\ = 12.92 A \end{aligned}$$

5. Power Dissipated Through Each Resistor.

or $P_1 = 125 \times 6.25 = 781.25 \text{ W}$ ~~781~~

~~$P_2 = 125 \times 1.25 = 156.25 \text{ W}$~~

~~$P_3 = 125 \times 2.5 = 312.5 \text{ W}$~~

a) $P_1 = 25 \times 1.25$
 $= 31.25 \text{ W}$

$P_2 = 37.5 \times 1.25$
 $= 46.875 \text{ W}$

$P_3 = 62.5 \times 1.25$
 $= 78.125 \text{ W}$

b) $P_1 = 125 \times 6.25$
 $= 781.25 \text{ W}$

$P_2 = 125 \times 1.25$
 $= 156.25 \text{ W}$

$P_3 = 125 \times 2.5$
 $= 312.5 \text{ W}$